

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Geochemistry L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus: Origin of elements-cosmological big bang, stellar evolution and nucleosynthesis, supernovae, s-, p-, r-processes; Chart of nuclides, distribution of nuclides and radioactive decay; Formation of solar nebula, nebular condensation of elements and compounds; Formation of terrestrial and Jovian planets; Planetary accretion and magma ocean formation; Meteorites-their classification, composition, comparison with terrestrial composition; Major and trace element geochemistry-classification of trace elements, behavior of trace elements; thermodynamic considerations of trace element solid solutions, trace element partitioning and fractionation during partial melting and crystallization of rocks and melts; Radioactive decay systems and principles of mass spectrometry; The origin and evolution of the crust, mantle and core using radiogenic and non-radiogenic isotope systems such as Rb-Sr, Sm-Nd, Lu-Hf, U-Th-Pb, and C, O, B, Li

Reference Books:

- 1) Cosmochemistry by McSween, H.Y. and Huss, G.R.
- 2) Isotopes-Principles and Applications by Gunter Faure and Teresa M. Mensing
- 3) Geochemistry by White, W.M

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Origin of elements - cosmological big bang, stellar evolution and nucleosynthesis, supernovae, s-, p-, r-processes; Chart of nuclides, distribution of nuclides and radioactive decay; Formation of solar nebula, nebular condensation of elements and compounds	9
2.	Formation of terrestrial and Jovian planets; Planetary accretion and magma ocean formation	2
3.	Meteorites-their classification, composition, comparison with terrestrial composition	4
4.	Major and trace element geochemistry-classification of trace elements, behavior of trace elements; thermodynamic considerations of trace element solid solutions, trace element partitioning and fractionation during partial melting and	9

	crystallization of rocks and melts	
5.	Radioactive decay systems and principles of mass spectrometry; The origin and evolution of the crust, mantle and core using radiogenic and non-radiogenic isotope systems such as Rb-Sr, Sm-Nd, Lu-Hf, U-Th-Pb, and C, O, B, Li	9
Total number of hours		33